

Question number	Answer	Additional guidance	Mark
7(a)	A description that includes the following point: - to be present in the blood (all the time) (1) and any TWO of the following: - needed to make thrombin (when blood needs to clot) (1) - which is an {enzyme / catalyst} (1) - so that fibrinogen can be converted into fibrin (1)	ACCEPT precursor of clotting process / inactive form of thrombin / inactive enzyme / inactive plasma protein NB thrombin catalyses fibrinogen into fibrin = 2 marks	(3)

Question number	Answer	Mark
7(b)	The only correct answer is A A anticoagulant <i>B is incorrect because antihypertensives treat high blood pressure</i> <i>C is incorrect because platelet inhibitors inhibit platelets, which are involved in the cascade before prothrombin and prothrombin is made by the liver</i> <i>D is incorrect because statins treat high blood cholesterol levels</i>	(1)

Question number	Answer	Additional guidance	Mark
7(c)(i)	An answer that includes the following points: - warfarin and vitamin K have a similar structure (1) - therefore warfarin {binds / blocks} to the {vitamin K epoxide reductase / VKOR} (1) (as a result of warfarin binding to enzyme) {less / no} vitamin K reduced (1)	ACCEPT both have rings / double bond oxygen ACCEPT warfarin is a {competitive / active-site directed } inhibitor / description DO NOT ACCEPT non-competitive inhibitor / description of one binds to vitamin K	(2)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	An answer that includes the following points: - increase in vitamin K would compete with warfarin for the active site (of vitamin K epoxide reductase / VKOR) (1) - therefore {some / more} vitamin K will be reduced (if vitamin K binds to enzyme) (1)	ACCEPT a description e.g. more enzyme substrate complexes	(2)

Question number	Answer	Additional guidance	Mark
7(d)	An explanation that includes the following points: • large groups of people (1) • (sample size is large) for reproducibility (1) • people in both groups consume the same {mass / volume / concentration} of vitamin K (1) • people in both groups consume the same concentration of drugs (1) • (variables controlled) for validity (1)	ACCEPT 20 + IGNORE reference to control groups ACCEPT repeatable / reliable IGNORE accurate / precise / valid IGNORE amount ACCEPT other appropriate named control variable e.g. sex, age, diet, level of activity, alcohol intake IGNORE same number of people in each group / amount IGNORE accurate / precise / reproducible / repeatable / reliable	(4)